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MANUFACTURING METROLOGY TEAM

Physics-Informed Machine Learning for Optical Surface Metrology: Bridging AI Research and Manufacturing

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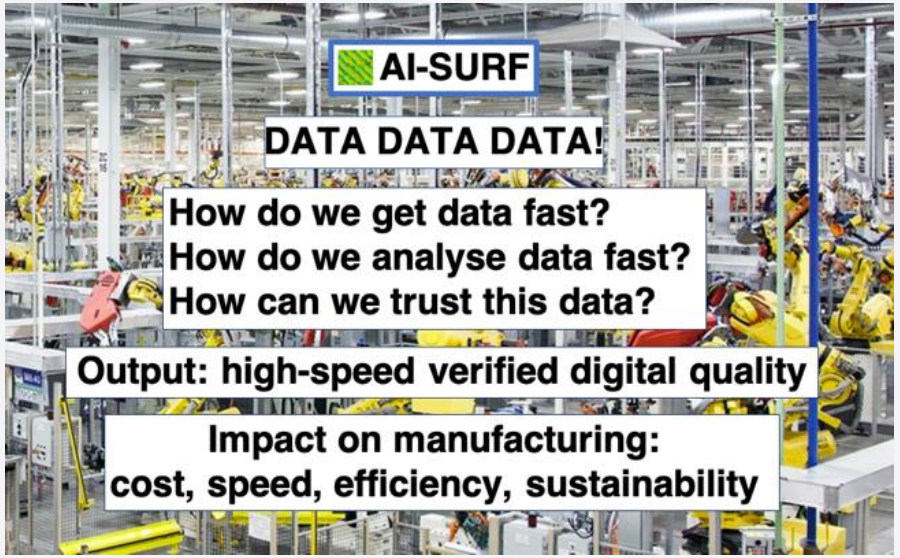
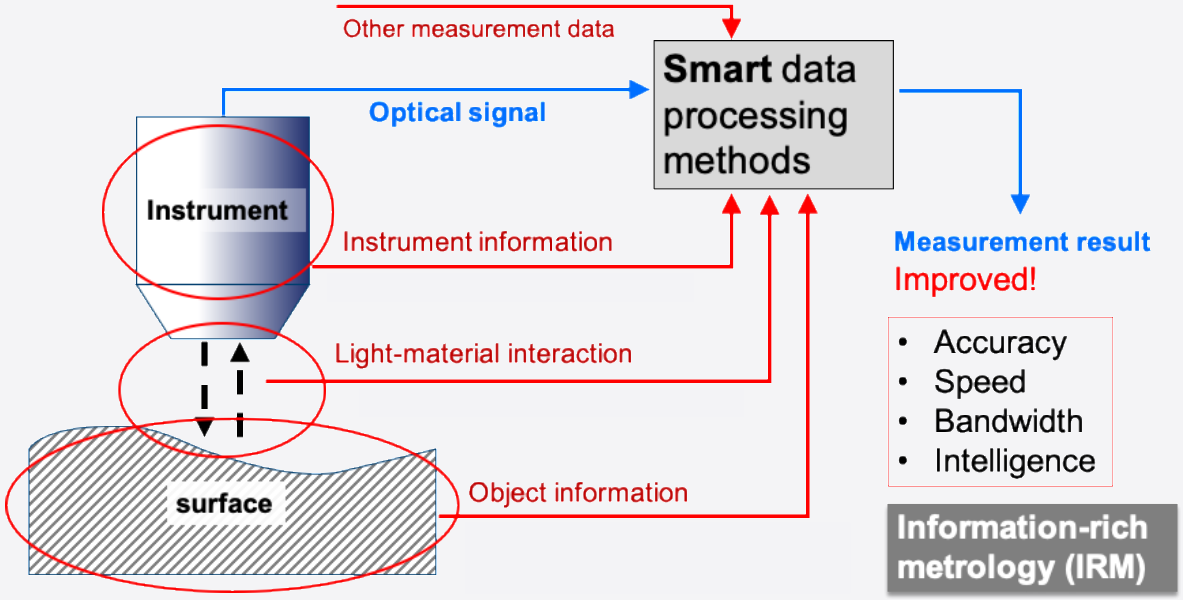
AI-SURF Project Motivation

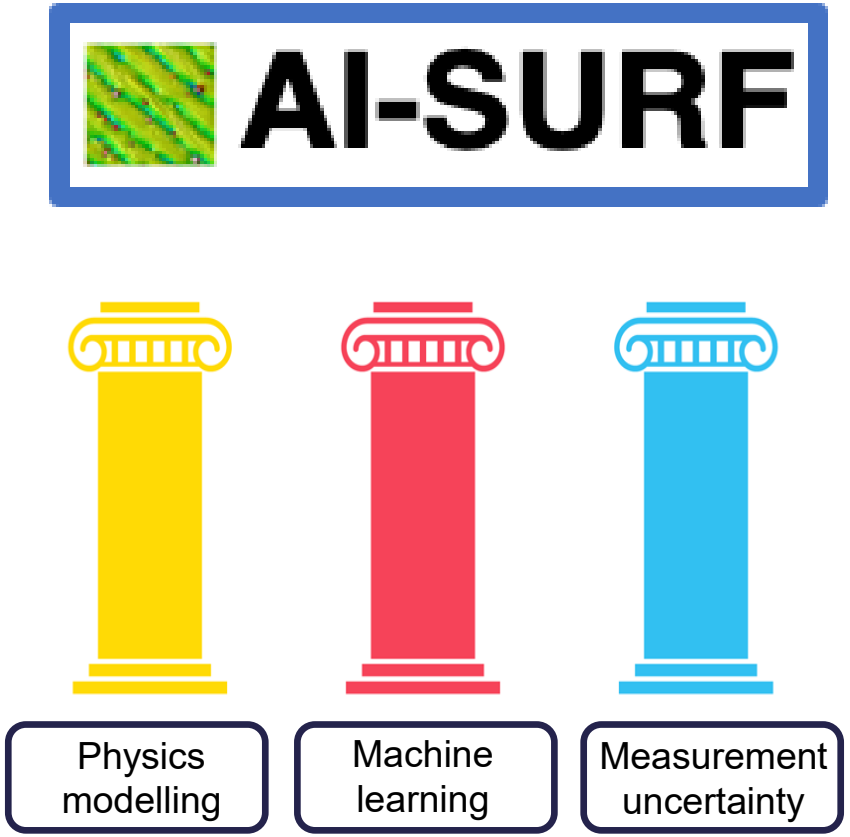
- In-process manufacturing quality control is stuck!
- Sensing and analysis methods are fundamentally too slow
- How can we ”break through” and address = AI-SURF!
- Why surfaces?
- Why optical technology?



AI-SURF Methodology

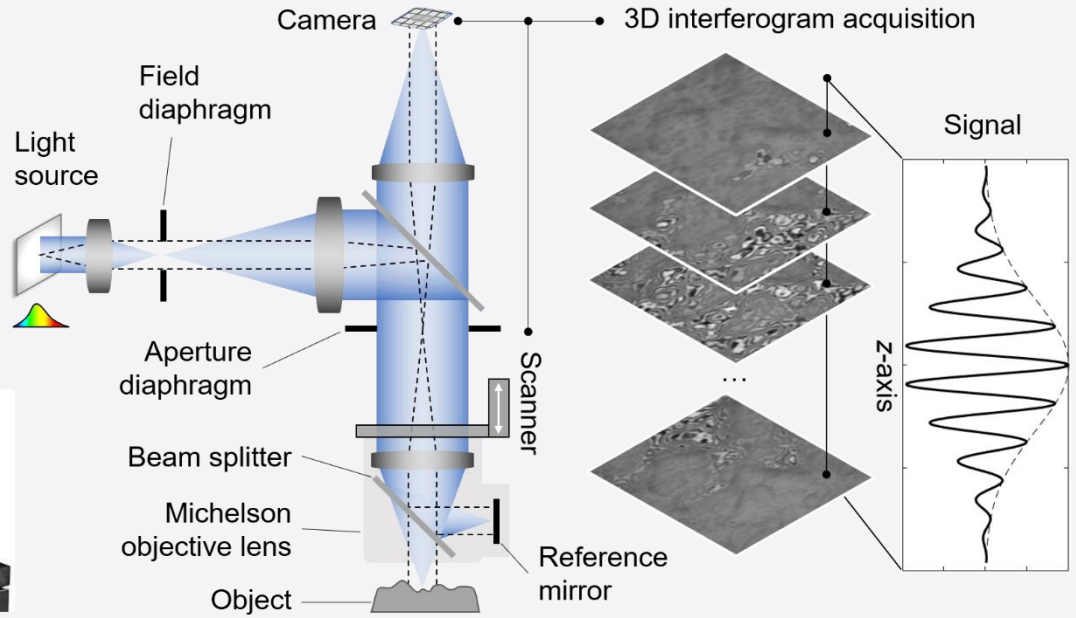
Information-rich metrology

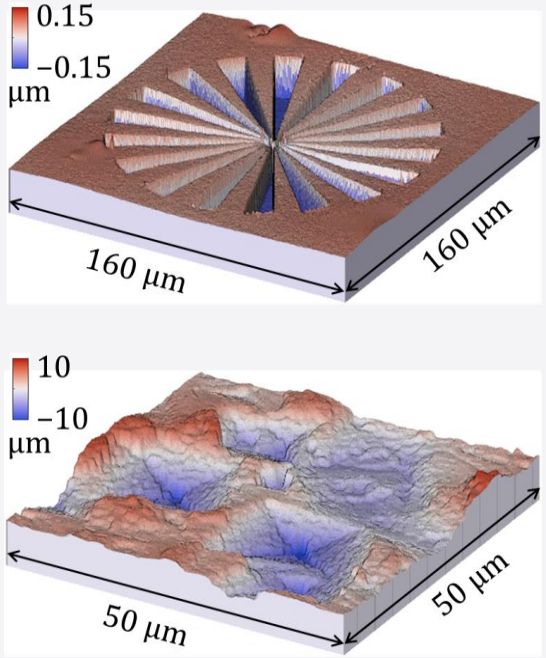




Outputs and state of the art boost

- **Software** to **predict** scattering from ***any*** **surface** geometry with ***any*** optical surface **instrument**
- **Machine learning** solutions to optical inverse problems

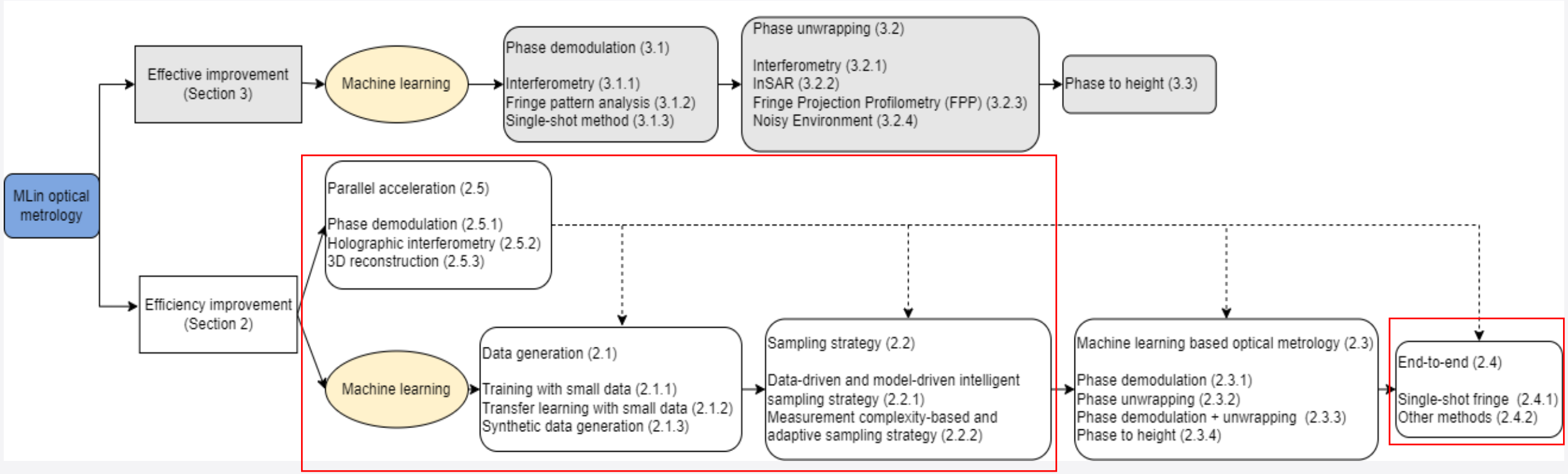




Su et al., Light: Advanced Manufacturing, 2, 2021.

Applying machine learning to optical metrology: a review

Structure of literature review



Ruidong, et al. Measurement Science and Technology (2024).

Why applying machine learning

Unlike **physics-based** approaches, machine learning has **data driven** nature

- Physics based models:
  - Mapping user desired sample parameters to simulate image
  - Investigate the influence of uncertainty sources
- Priori information enhanced user dependent models:
  - Intricate and time consuming
- Multi-step measurement processes:
  - Error accumulation
- Sensitivity of optical measurement to the environment:
  - Laboratory settings
  - Harsh factory environments due to physical and knowledge limitations

When applying machine learning

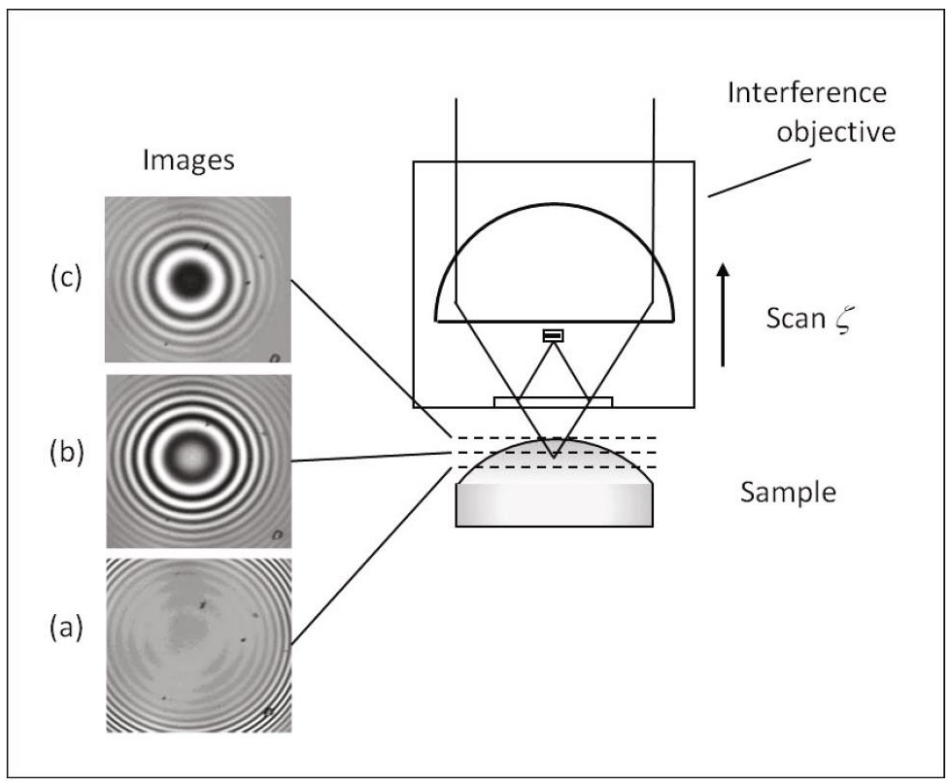
- Machine learning not always has correct solutions, while **traceability, reliability, and repeatability** are prioritised
- **Unexplainability** and incomprehensibility of machine learning. **Noisy, inaccurate, uncertain, vague, and incomplete datasets** of optical metrology
- Machine learning relies on **massive** training datasets without significant reliance on **priori knowledge**
- Machine learning depends on the **reliability** of data. Few **publicly** available datasets, hindering fair and standardised algorithm comparisons

Limitation of machine learning

- The “**black box**” nature of machine learning, may not always adhere strictly to the physical laws that govern the traditional models. This can sometimes lead to results that are not **physically plausible**, thereby affecting the **reliability and robustness** of the system, especially under conditions not represented in the training data
  - **Hybrid models** that combine **ML** with **physics-based** models
  - Designing an **interpretable evaluation** system

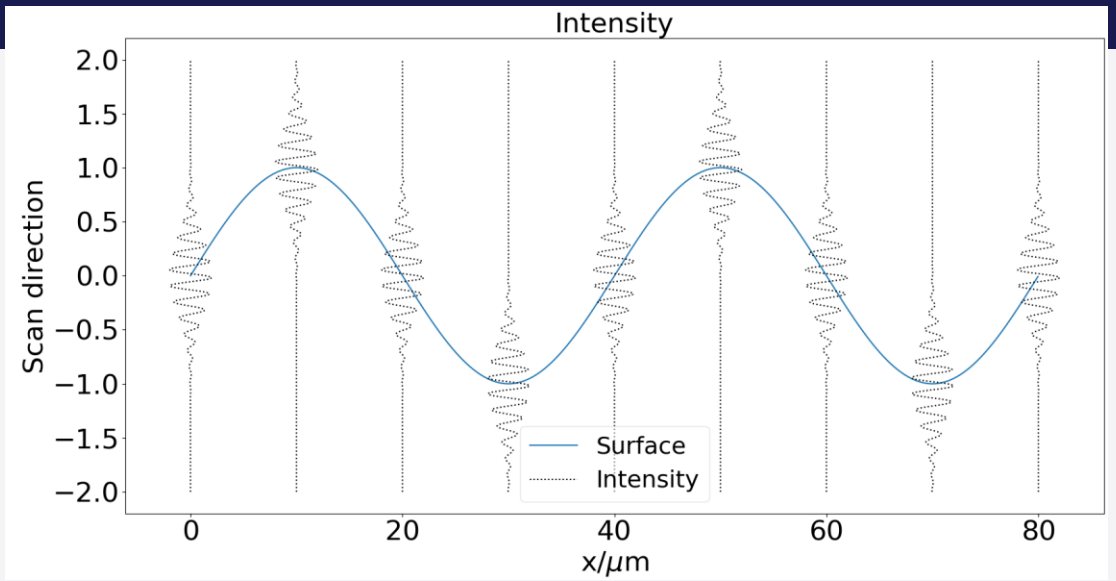
- ML are dependent on the **quality and range** of the training data. In real-world, where **data anomalies or novel conditions** outside of the training set can occur, the ML system may fail to generalize effectively

Introduction of coherence scanning interferometry (CSI)

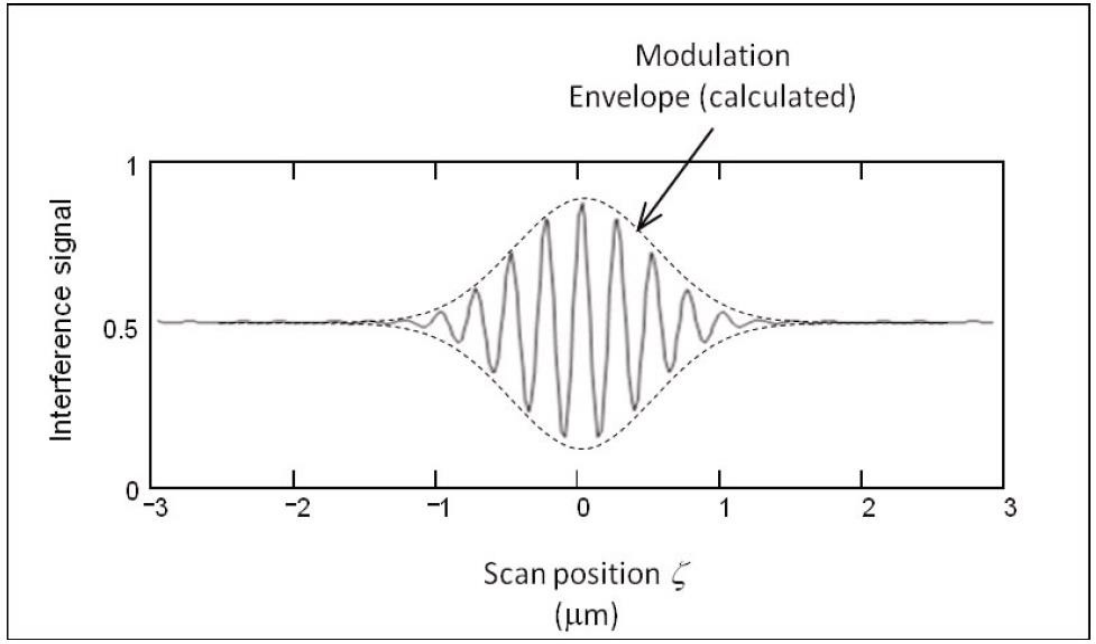


a [1] Images of interference fringes on a curved surface with low coherence illumination

b Example of scanning process



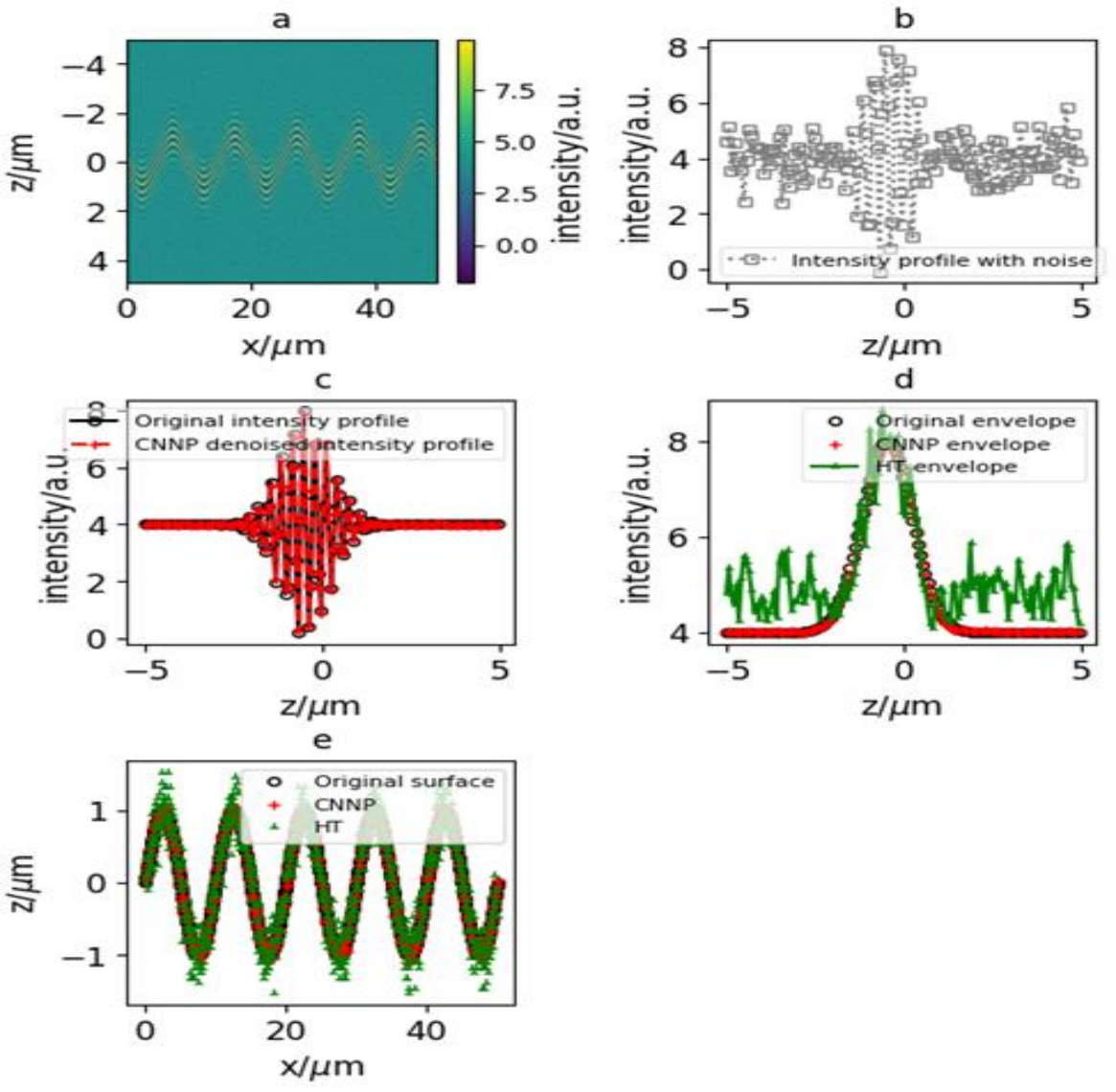
c [1] CSI signal for a single pixel showing the modulation envelope



- [1] de Groot, P. (2011). Coherence Scanning Interferometry. In: Leach, R. (eds) Optical Measurement of Surface Topography. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-12012-1_9

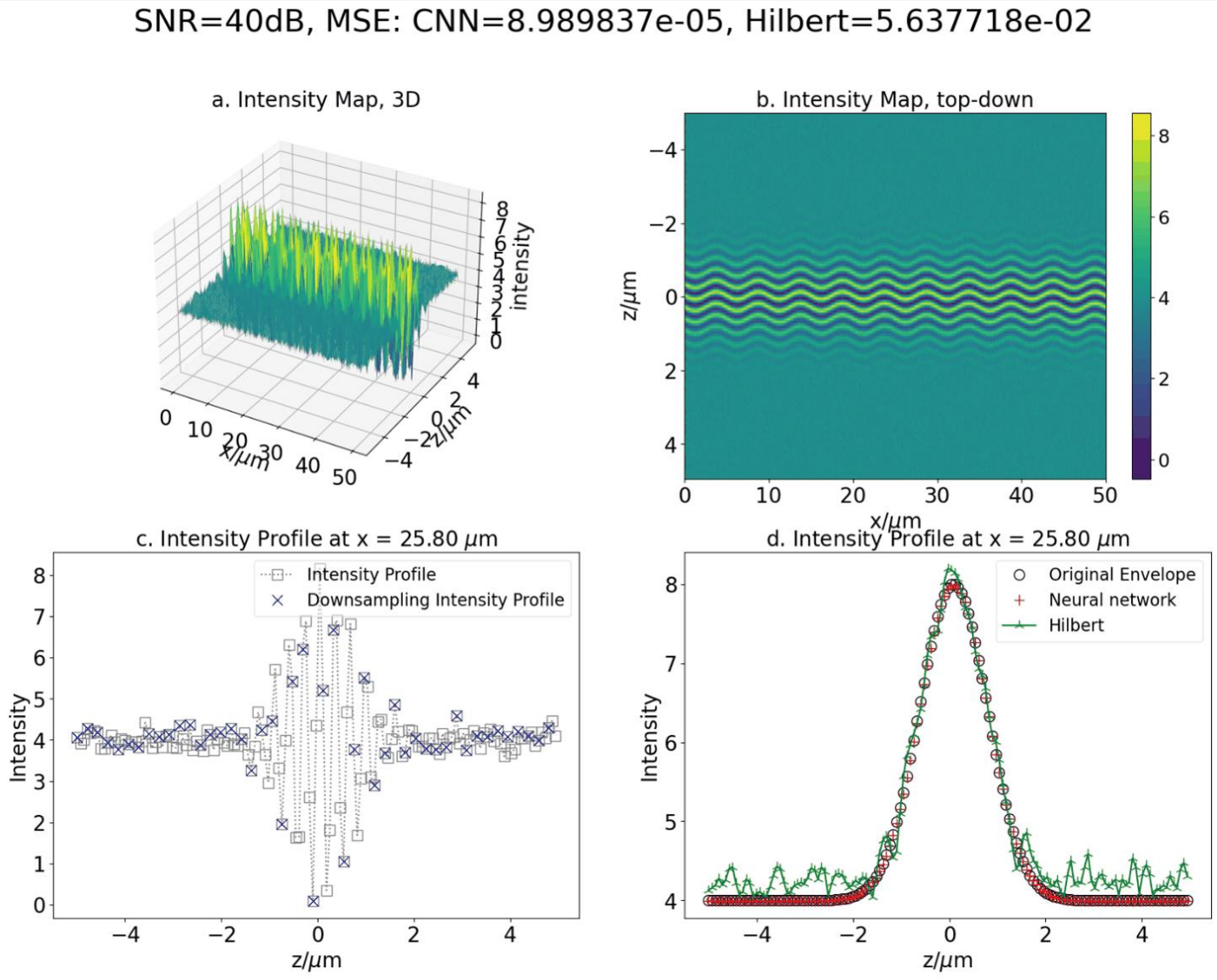
Experiment

- Introduce a software to **enhance noise reduction in CSI**
- Use machine learning and physical modelling principles to achieve better performance in surface reconstruction
- **CNNP reduces** mean squared errors (MSE) by an order of magnitude **compared to HT**
- CNNP MSE: 5.015×10^{-3} µm², HT MSE: 5.219×10^{-2} µm²
- Figure shows the **denoising** process on simulated CSI signal on sinusoidal profile with added noise



Experiment: Noise Reduction

- SNR 40 to 1dB
- CNN VS Hilbert



Conclusion

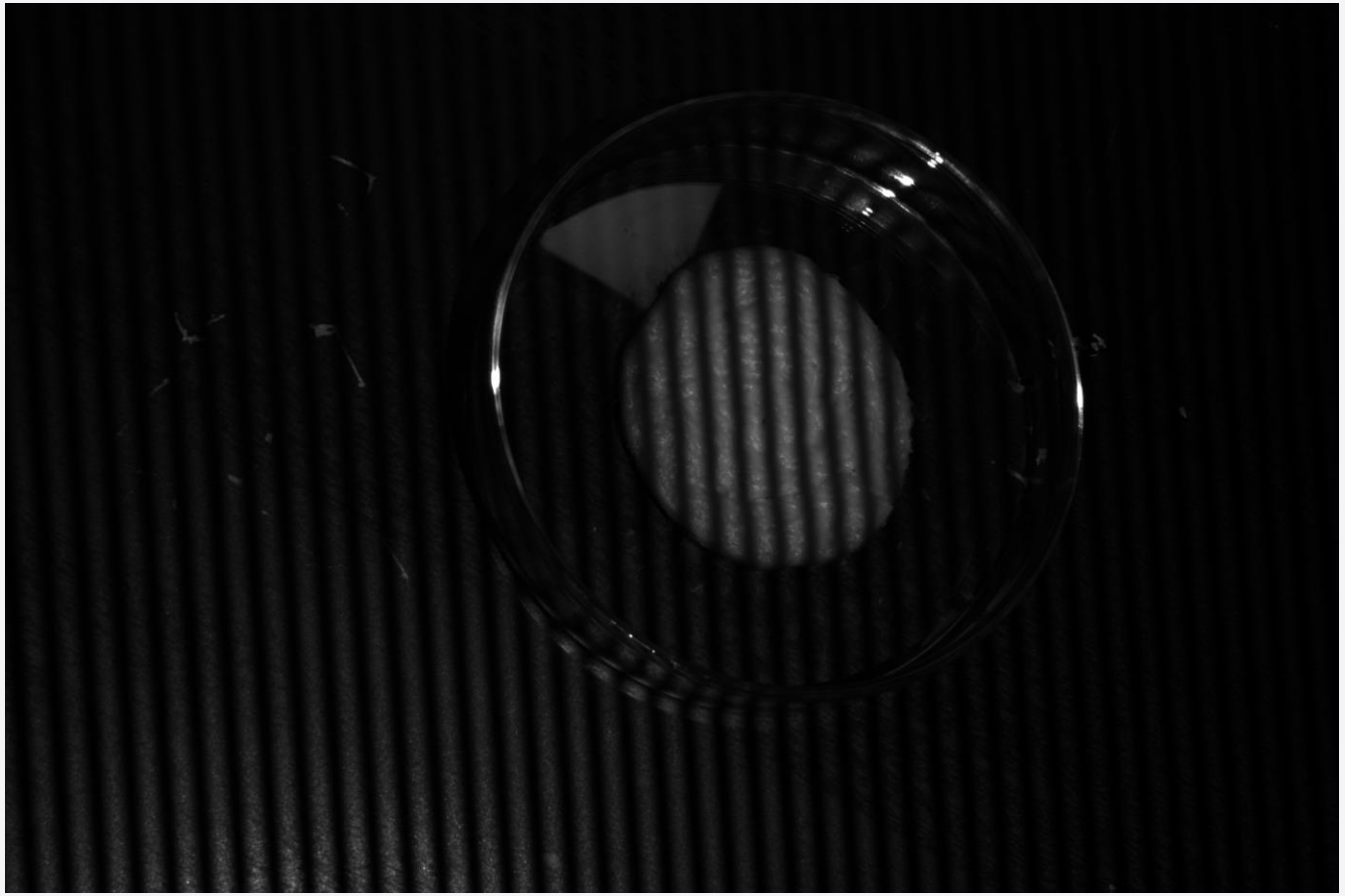
- We introduced a **CNNP** that combines **ML** with the **HT** method to improve the **robustness and accuracy** of surface reconstruction from **noisy** data.
- Our findings demonstrate that the **integration of ML with the HT-envelope** method enhances the accuracy of surface reconstruction in CSI under noisy.
- The combination of CNNs with the HT method reduces the impact of environmental noise and low SNRs, crucial for **applications** requiring precise surface measurements in **uncontrolled environments**

Calibration Optimisation of Fringe Projection Profilometry by Using a Virtual Instrument

The Challenge

Traditional FPP Calibration Limitations

- **Current Issues:**
- Time-consuming physical calibration procedures
- Limited ability to test edge cases
- Difficulty in isolating error sources
- **Impact:**
- Calibration time: Time consuming per setup
- Success rate: Variable depending on operator skill



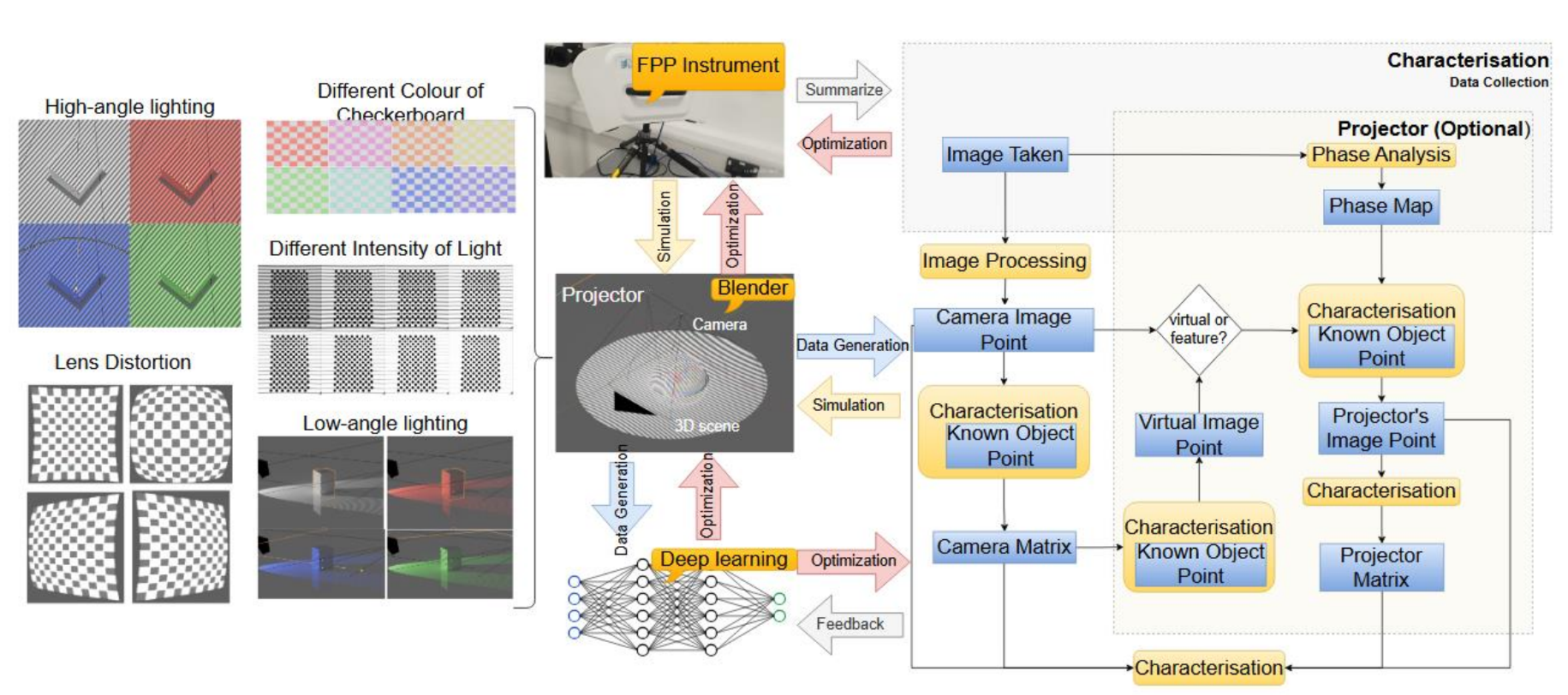
Research Question

Can a Virtual Instrument Accelerate and Improve FPP Charact?

- **Virtual Instrument enables:**
- Rapid parameter optimization before physical implementation
- Perfect ground truth for algorithm validation
- Reduced calibration iterations

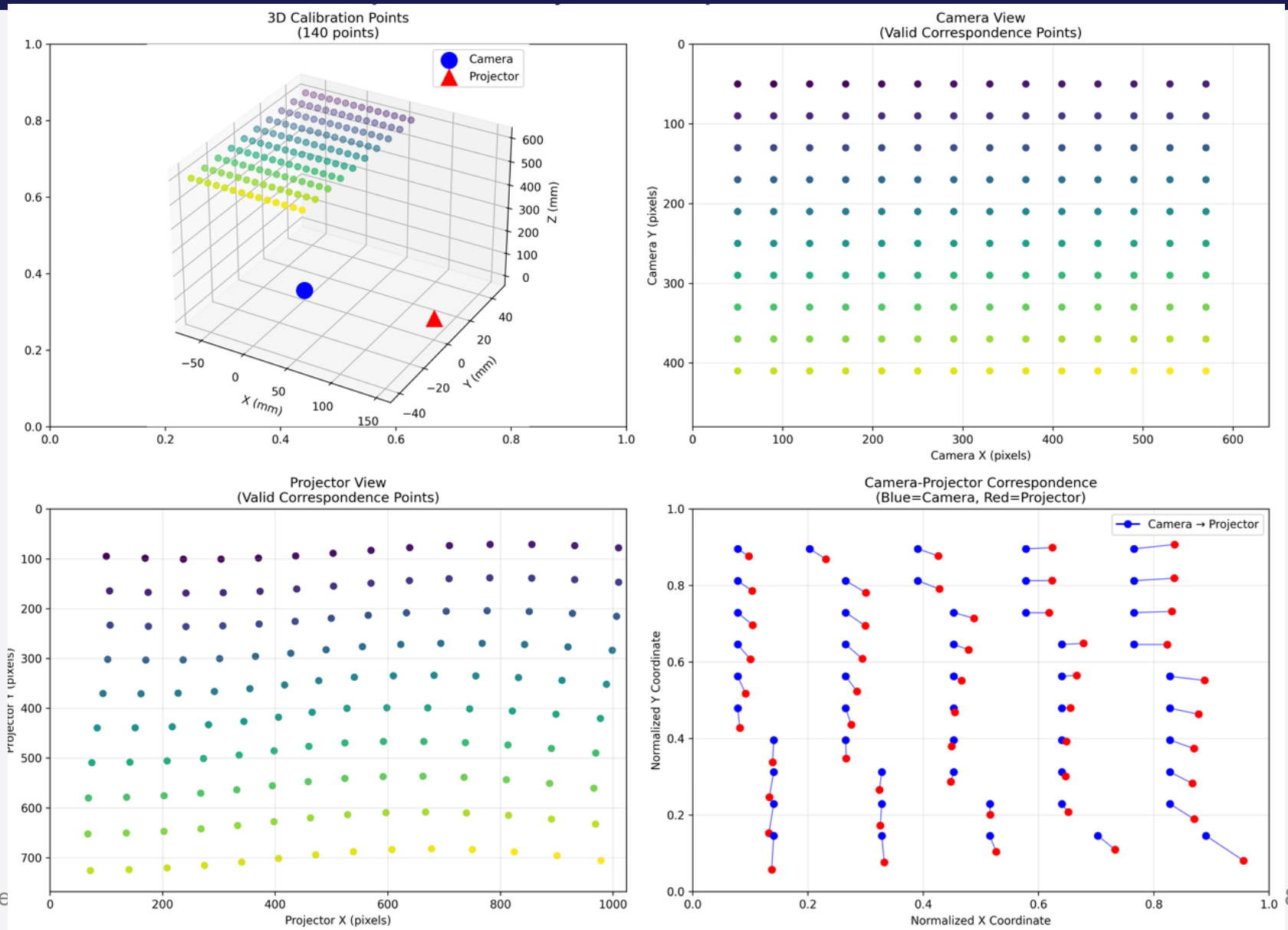
Software: Virtual Instrument Architecture

CheckerBoard	Camera	Projector
Size	Size	size
Type	Position	Position
Colour	Rotation	Rotation
Position	Number	Number
Rotation	Field of View	Field of View
Number	Focal length	Focal length
	Resolution	Resolution
	Distortion	Distortion
	Lens shift	Light Intensity
		Light colour
	...	...

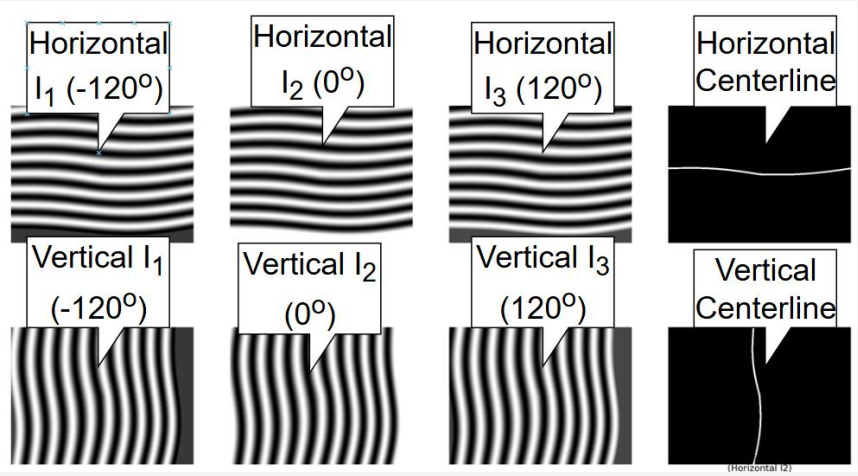


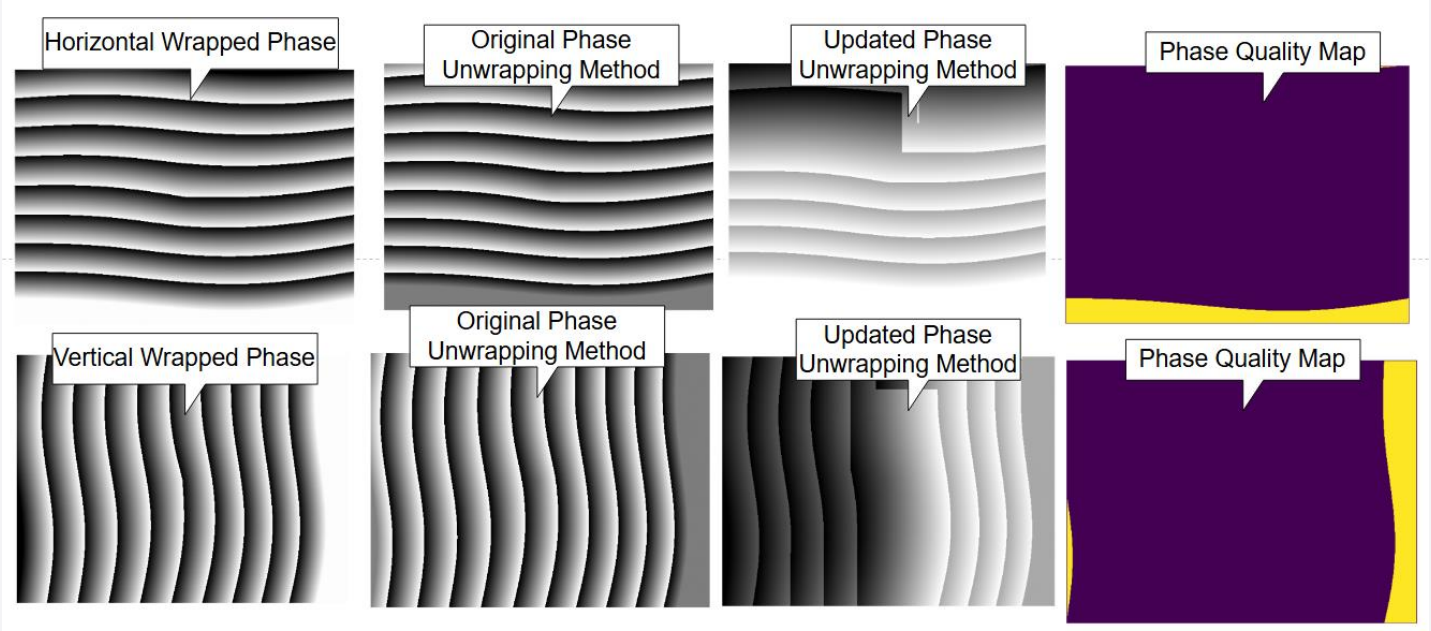
Projection & Correspondence

- Perspective projection maps 3D world points onto 2D camera and projector pixels.
- Calibration points projected into both devices define a one to one correspondence.
- Normalised coordinate plots illustrate how the geometric mapping warps from 3D to 2D.



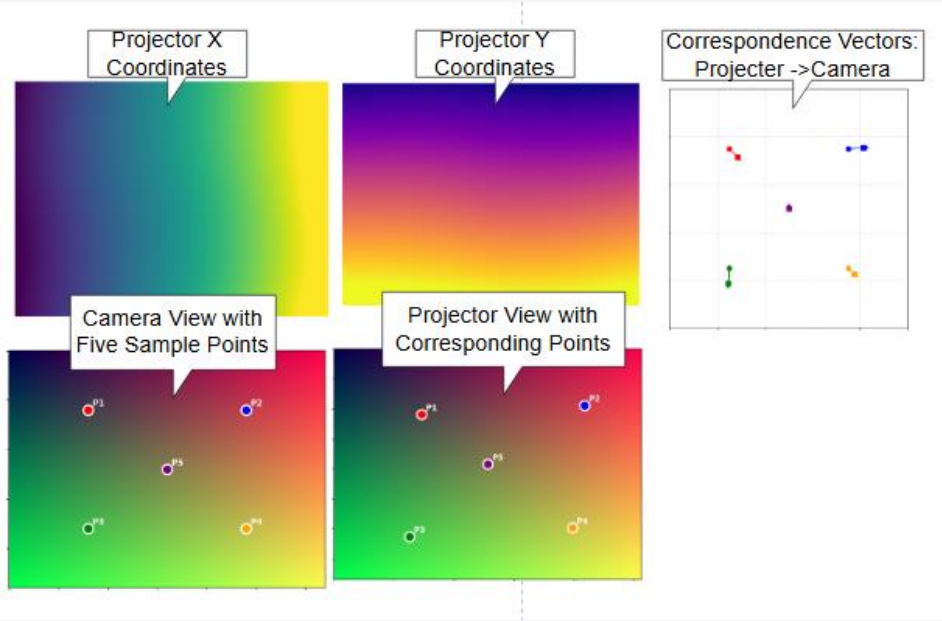
Fringe Generation & Phase Computation





- Generate sinusoidal fringe patterns at multiple phase shifts and project them along horizontal and vertical directions.
- Compute wrapped phase from the three fringe intensities; derive absolute phase by referencing a bright centreline.
- Analyse fringe profiles to verify phase variation and intensity modulation.
- 2D unwrapping resolves discontinuities and produces continuous absolute phase maps.

Final Mapping



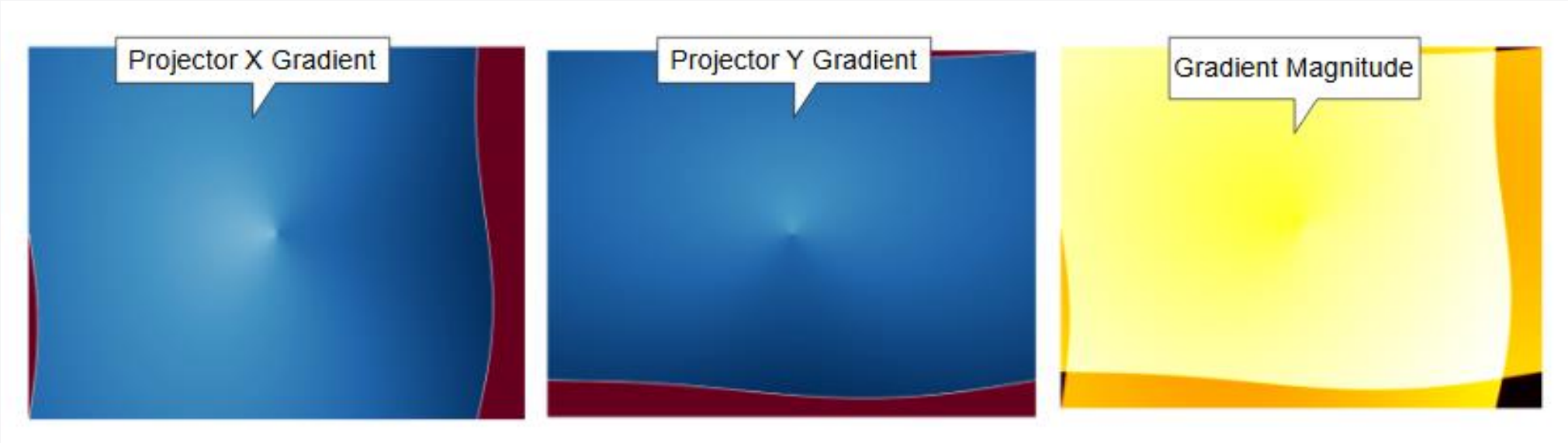
Point	Camera (X,Y)	Projector (X,Y)	Vector (ΔX,ΔY)
P1	(160, 120)	(291.4, 218.5)	(131.4, 98.5)
P2	(480, 120)	(834.1, 188.5)	(354.1, 68.5)
P3	(160, 360)	(251.4, 625.6)	(91.4, 265.6)
P4	(480, 360)	(794.1, 595.6)	(314.1, 235.6)
P5	(320, 240)	(512.0, 384.0)	(192.0, 144.0)

- Final correspondence maps demonstrate smooth mapping from camera to projector without overlaps.

Camera View

Projector View

Calibration Accuracy & Quality



- Compute gradients of projector coordinates to assess smoothness of the correspondence mapping.
- High gradient magnitudes indicate abrupt mapping variations; evaluate distribution of gradients.
- Quality maps guide unwrapping and highlight valid calibration regions on the plane.
- This method yelds precise sub-pixel correspondences and decoupled calibration.

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Thank you

EMPIR EURAMET

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MIDLANDS CENTRE For Data-Driven Metrology

AI-SURF

UKRI Engineering and Physical Sciences Research Council